Jul 31 2025
BDBP106: Linux and Python programming

Lab 4
Learning goals: Linux OS commands ln, file, cat, more, less, head, tail, history and sort

NOTE: Save screenshots of each exercise, and upload your work to your github account as Lab4.pdf by end of Thursday Jul 31.
For this lab, download 'Heart.csv' to a folder called 'Lab4'. If the file download to the "Downloads" folder by default, move the file to the folder 'Lab4'.

EXERCISES Start a fresh terminal for this Lab.

(1) Create symbolic and hard links to the data file you downloaded. What happens if you use the same name for the links? Print the screenshot of the output message and paste it in your answers. Also how do you prove that you have created the links correctly? Use the appropriate command to prove it.

Hard link: Hardheart
Soft link: symheart
If you use the same name for the links the output shows : "file exists"
Command used to check if links are created: ls -li
ls provides the listing while ls -li provides the inode number aswell. The inode number for hard link and original file are same.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ ln Heart Hardheart
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ ln -s Heart
ln: failed to create symbolic link './Heart': File exists
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ ln -s symheart
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ ls
etc_learn  etc.zip  file1  file2  file3  file3_sorted  Hardheart  Heart  lab2.pdf  lab4.pdf  symheart
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ ln Heart Heart
ln: failed to create hard link 'Heart': File exists
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ ls -li
total 264
28594288 drwxrwxrwx 83 ibab ibab  4096 Jul 22 16:59 etc_learn
28596577 -rw-rw-r--  1 ibab ibab 96755 Jul 24 14:34 etc.zip
28603737 -rw-rw-r--  1 ibab ibab    24 Jul 31 09:30 file1
28603773 -rw-rw-r--  1 ibab ibab    23 Jul 31 09:31 file2
28603777 -rw-rw-r--  1 ibab ibab    48 Jul 31 09:31 file3
28603918 -rw-rw-r--  1 ibab ibab    48 Jul 31 09:39 file3_sorted
28577417 -rw-rw-r--  3 ibab ibab 19925 Jul 31 14:00 Hardheart
28577417 -rw-rw-r--  3 ibab ibab 19925 Jul 31 14:00 Heart
28588328 -rw-rw-r--  1 ibab ibab 47489 Jul 18 15:04 lab2.pdf
28577365 -rw-rw-r--  1 ibab ibab 57441 Jul 31 14:00 lab4.pdf
28577987 lrwxrwxrwx  1 ibab ibab     8 Jul 31 14:25 symheart -> symheart
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$
```

(2) What command will you use to determine the filetype of the downloaded data file? Execute this command and take a screenshot of the command and the output to paste it in your work.
Command used to determine the filetype: file <filename>
Output

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ file Heart
Heart: CSV text
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$
```

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(3) View the contents of the data file using more and less commands. How many pages are there?
More command: 1
Less command: 5

(4) Output the first 35 lines of the data file using the appropriate command. Save the screenshot in your work.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ head -n 35 Heart
"","Age","Sex","ChestPain","RestBP","Chol","Fbs","RestECG","MaxHR","ExAng","Oldpeak","Slope","Ca","Thal","AHD"
"1",63,1,"typical",145,233,1,2,150,0,2.3,3,0,"fixed","No"
"2",67,1,"asymptomatic",160,286,0,2,108,1,1.5,2,3,"normal","Yes"
"3",67,1,"asymptomatic",120,229,0,2,129,1,2.6,2,2,"reversable","Yes"
"4",37,1,"nonanginal",130,250,0,0,187,0,3.5,3,0,"normal","No"
"5",41,0,"nontypical",130,204,0,2,172,0,1.4,1,0,"normal","No"
"6",56,1,"nontypical",120,236,0,0,178,0,0.8,1,0,"normal","No"
"7",62,0,"asymptomatic",140,268,0,2,160,0,3.6,3,2,"normal","Yes"
"8",57,0,"asymptomatic",120,354,0,0,163,1,0.6,1,0,"normal","No"
"9",63,1,"asymptomatic",130,254,0,2,147,0,1.4,2,1,"reversable","Yes"
"10",53,1,"asymptomatic",140,203,1,2,155,1,3.1,3,0,"reversable","Yes"
"11",57,1,"asymptomatic",140,192,0,0,148,0,0.4,2,0,"fixed","No"
"12",56,0,"nontypical",140,294,0,2,153,0,1.3,2,0,"normal","No"
"13",56,1,"nonanginal",130,256,1,2,142,1,0.6,2,1,"fixed","Yes"
"14",44,1,"nontypical",120,263,0,0,173,0,0,1,0,"reversable","No"
"15",52,1,"nonanginal",172,199,1,0,162,0,0.5,1,0,"reversable","No"
"16",57,1,"nonanginal",150,168,0,0,174,0,1.6,1,0,"normal","No"
"17",48,1,"nontypical",110,229,0,0,168,0,1,3,0,"reversable","Yes"
"18",54,1,"asymptomatic",140,239,0,0,160,0,1.2,1,0,"normal","No"
"19",48,0,"nonanginal",130,275,0,0,139,0,0.2,1,0,"normal","No"
"20",49,1,"nontypical",130,266,0,0,171,0,0.6,1,0,"normal","No"
"21",64,1,"typical",110,211,0,2,144,1,1.8,2,0,"normal","No"
"22",58,0,"typical",150,283,1,2,162,0,1,1,0,"normal","No"
"23",58,1,"nontypical",120,284,0,2,160,0,1.8,2,0,"normal","Yes"
"24",58,1,"nonanginal",132,224,0,2,173,0,3.2,1,2,"reversable","Yes"
"25",60,1,"asymptomatic",130,206,0,2,132,1,2.4,2,2,"reversable","Yes"
"26",50,0,"nonanginal",120,219,0,0,158,0,1.6,2,0,"normal","No"
"27",58,0,"nonanginal",120,340,0,0,172,0,0,1,0,"normal","No"
"28",66,0,"typical",150,226,0,0,114,0,2.6,3,0,"normal","No"
"29",43,1,"asymptomatic",150,247,0,0,171,0,1.5,1,0,"normal","No"
"30",40,1,"asymptomatic",110,167,0,2,114,1,2,2,0,"reversable","Yes"
"31",69,0,"typical",140,239,0,0,151,0,1.8,1,2,"normal","No"
"32",60,1,"asymptomatic",117,230,1,0,160,1,1.4,1,2,"reversable","Yes"
"33",64,1,"nonanginal",140,335,0,0,158,0,0,1,0,"normal","Yes"
"34",59,1,"asymptomatic",135,234,0,0,161,0,0.5,2,0,"reversable","No"
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$
```

(5) Output the last 15 lines of the data file using the appropriate command. Save the screenshot in your work.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ tail -n 15 Heart
"289",56,1,"nontypical",130,221,0,2,163,0,0,1,0,"reversable","No"
"290",56,1,"nontypical",120,240,0,0,169,0,0,3,0,"normal","No"
"291",67,1,"nonanginal",152,212,0,2,150,0,0.8,2,0,"reversable","Yes"
"292",55,0,"nontypical",132,342,0,0,166,0,1.2,1,0,"normal","No"
"293",44,1,"asymptomatic",120,169,0,0,144,1,2.8,3,0,"fixed","Yes"
"294",63,1,"asymptomatic",140,187,0,2,144,1,4,1,2,"reversable","Yes"
"295",63,0,"asymptomatic",124,197,0,0,136,1,0,2,0,"normal","Yes"
"296",41,1,"nontypical",120,157,0,0,182,0,0,1,0,"normal","No"
"297",59,1,"asymptomatic",164,176,1,2,90,0,1,2,2,"fixed","Yes"
"298",57,0,"asymptomatic",140,241,0,0,123,1,0.2,2,0,"reversable","Yes"
"299",45,1,"typical",110,264,0,0,132,0,1.2,2,0,"reversable","Yes"
"300",68,1,"asymptomatic",144,193,1,0,141,0,3.4,2,2,"reversable","Yes"
"301",57,1,"asymptomatic",130,131,0,0,115,1,1.2,2,1,"reversable","Yes"
"302",57,0,"nontypical",130,236,0,2,174,0,0,2,1,"normal","Yes"
"303",38,1,"nonanginal",138,175,0,0,173,0,0,1,NA,"normal","No"
```

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(6) Use the history command to list the last few commands, and execute the second last command in your list. Do this using both the process ID and the first letter of the command.

```
  415  ln -s Heart
  416  ln -s symheart
  417  ls
  418  ln Heart Heart
  419  ls -li
  420  file Heart
  421  more Heart
  422  less Heart
  423  tail -n 15 Heart
  424  head -n 35 Heart
  425  history
  426  history h
  427  history
  428  pwd
  429  history
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ history !428
history pwd
bash: history: pwd: numeric argument required
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ history !p
history pwd
bash: history: pwd: numeric argument required
```

(7) The sort command. For each of the exercises below, find the correct combination of options for the sort command using the man pages. Make sure to show your sequence of steps clearly in your submission work.
(i) Sort the data according to the first column, keeping in mind that the first column has numbers. Save the output in a new file called col1_sorted.out.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ sort -t ',' -V -k 1 Heart -o col1_sorted.out
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ cat col1_sorted.out
"1",63,1,"typical",145,233,1,2,150,0,2.3,3,0,"fixed","No"
"2",67,1,"asymptomatic",160,286,0,2,108,1,1.5,2,3,"normal","Yes"
"3",67,1,"asymptomatic",120,229,0,2,129,1,2.6,2,2,"reversable","Yes"
"4",37,1,"nonanginal",130,250,0,0,187,0,3.5,3,0,"normal","No"
"5",41,0,"nontypical",130,204,0,2,172,0,1.4,1,0,"normal","No"
"6",56,1,"nontypical",120,236,0,0,178,0,0.8,1,0,"normal","No"
"7",62,0,"asymptomatic",140,268,0,2,160,0,3.6,3,2,"normal","Yes"
"8",57,0,"asymptomatic",120,354,0,0,163,1,0.6,1,0,"normal","No"
"9",63,1,"asymptomatic",130,254,0,2,147,0,1.4,2,1,"reversable","Yes"
"10",53,1,"asymptomatic",140,203,1,2,155,1,3.1,3,0,"reversable","Yes"
"11",57,1,"asymptomatic",140,192,0,0,148,0,0.4,2,0,"fixed","No"
"12",56,0,"nontypical",140,294,0,2,153,0,1.3,2,0,"normal","No"
"13",56,1,"nonanginal",130,256,1,2,142,1,0.6,2,1,"fixed","Yes"
"14",44,1,"nontypical",120,263,0,0,173,0,0,1,0,"reversable","No"
"15",52,1,"nonanginal",172,199,1,0,162,0,0.5,1,0,"reversable","No"
"16",57,1,"nonanginal",150,168,0,0,174,0,1.6,1,0,"normal","No"
"17",48,1,"nontypical",110,229,0,0,168,0,1,3,0,"reversable","Yes"
"18",54,1,"asymptomatic",140,239,0,0,160,0,1.2,1,0,"normal","No"
"19",48,0,"nonanginal",130,275,0,0,139,0,0.2,1,0,"normal","No"
```

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(ii) Sort the data according to the ‘Age’ column. Save the output in a new file called age_sorted.out.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ sort -t ',' -V -k2 Heart -o age_sorted.out
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ cat age_sorted.out
"133",29,1,"nontypical",130,204,0,2,202,0,0,1,0,"normal","No"
"226",34,0,"nontypical",118,210,0,0,192,0,0.7,1,0,"normal","No"
"102",34,1,"typical",118,182,0,2,174,0,0,1,0,"normal","No"
"118",35,0,"asymptomatic",138,183,0,0,182,0,1.4,1,0,"normal","No"
"139",35,1,"asymptomatic",120,198,0,0,130,1,1.6,2,0,"reversable","Yes"
"169",35,1,"asymptomatic",126,282,0,2,156,1,0,1,0,"reversable","Yes"
"284",35,1,"nontypical",122,192,0,0,174,0,0,1,0,"normal","No"
"211",37,0,"nonanginal",120,215,0,0,170,0,0,1,0,"normal","No"
"4",37,1,"nonanginal",130,250,0,0,187,0,3.5,3,0,"normal","No"
"303",38,1,"nonanginal",138,175,0,0,173,0,0,1,NA,"normal","No"
"212",38,1,"typical",120,231,0,0,182,1,3.8,2,0,"reversable","Yes"
"223",39,0,"nonanginal",94,199,0,0,179,0,0,1,0,"normal","No"
"278",39,0,"nonanginal",138,220,0,0,152,0,0,2,0,"normal","No"
"110",39,1,"asymptomatic",118,219,0,0,140,0,1.2,2,0,"reversable","Yes"
"83",39,1,"nonanginal",140,321,0,2,182,0,0,1,0,"normal","No"
"30",40,1,"asymptomatic",110,167,0,2,114,1,2,2,0,"reversable","Yes"
"269",40,1,"asymptomatic",152,223,0,0,181,0,0,1,0,"reversable","Yes"
"42",40,1,"typical",140,199,0,0,178,1,1.4,1,0,"reversable","No"
"221",41,0,"nonanginal",112,268,0,2,172,1,0,1,0,"normal","No"
"51",41,0,"nontypical",105,198,0,0,168,0,0,1,1,"normal","No"
"242",41,0,"nontypical",126,306,0,0,163,0,0,1,0,"normal","No"
"5",41,0,"nontypical",130,204,0,2,172,0,1.4,1,0,"normal","No"
"58",41,1,"asymptomatic",110,172,0,2,158,0,0,1,0,"reversable","Yes"
"148",41,1,"nonanginal",112,250,0,0,179,0,0,1,0,"normal","No"
"213",41,1,"nonanginal",130,214,0,2,168,0,2,2,0,"normal","No"
"241",41,1,"nontypical",110,235,0,0,153,0,0,1,0,"normal","No"
"296",41,1,"nontypical",120,157,0,0,182,0,0,1,0,"normal","No"
"116",41,1,"nontypical",135,203,0,0,132,0,0,2,0,"fixed","No"
"152",42,0,"asymptomatic",102,265,0,2,122,0,0.6,2,0,"normal","No"
"256",42,0,"nonanginal",120,209,0,0,173,0,0,2,0,"normal","No"
"266",42,1,"asymptomatic",136,315,0,0,125,1,1.8,2,0,"fixed","Yes"
"36",42,1,"asymptomatic",140,226,0,0,178,0,0,1,0,"normal","No"
"187",42,1,"nonanginal",120,240,1,0,194,0,0.8,3,0,"reversable","No"
"270",42,1,"nonanginal",130,180,0,0,150,0,0,1,0,"normal","No"
"240",42,1,"nontypical",120,295,0,0,162,0,0,1,0,"normal","No"
"183",42,1,"typical",148,244,0,2,178,0,0.8,1,2,"normal","No"
"114",43,0,"asymptomatic",132,341,1,2,136,1,3,2,0,"reversable","Yes"
"135",43,0,"nonanginal",122,213,0,0,165,0,0.2,2,0,"normal","No"
```

(iii) Sort the data in a reverse manner according to the ’RestBP’ column. Save the output in a new file called restbp_revsort.out.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ sort -t ',' -V -r -k 5 Heart -o restbp_revsort.out
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ cat restbp_revsort.out
"","Age","Sex","ChestPain","RestBP","Chol","Fbs","RestECG","MaxHR","ExAng","Oldpeak","Slope","Ca","Thal","AHD"
"127",56,0,"asymptomatic",200,288,1,2,133,1,4,3,2,"reversable","Yes"
"189",54,1,"nontypical",192,283,0,2,195,0,0,1,1,"reversable","Yes"
"232",55,0,"asymptomatic",180,327,0,1,117,1,3.4,2,0,"normal","Yes"
"202",64,0,"asymptomatic",180,325,0,0,154,1,0,1,0,"normal","No"
"84",68,1,"nonanginal",180,274,1,2,150,1,1.6,2,0,"reversable","Yes"
"184",59,1,"typical",178,270,0,2,145,0,4.2,3,0,"reversable","No"
"214",66,0,"asymptomatic",178,228,1,0,165,1,1,2,2,"reversable","Yes"
```

(iv) Repeat (ii) and (iii) by giving GNU-style parameters to the sort command. This is where man pages are extremely useful!

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ sort --field-separator=',' --version-sort --key=2 Heart --output=age_sorted.out
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ sort --field-separator=',' --version-sort --reverse --key=5 Heart --output=restbp_revsort.out
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$
```

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(v) Sort the data first according to age, and then according to RestBP. How would you do this? Explain the steps clearly.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ sort -t ',' -k2,2 -k5,5 -n Heart
"","Age","Sex","ChestPain","RestBP","Chol","Fbs","RestECG","MaxHR","ExAng","Oldpeak","Slope","Ca","Thal","AHD"
"133",29,1,"nontypical",130,204,0,2,202,0,0,1,0,"normal","No"
"102",34,1,"typical",118,182,0,2,174,0,0,1,0,"normal","No"
"226",34,0,"nontypical",118,210,0,0,192,0,0.7,1,0,"normal","No"
"139",35,1,"asymptomatic",120,198,0,0,130,1,1.6,2,0,"reversable","Yes"
"284",35,1,"nontypical",122,192,0,0,174,0,0,1,0,"normal","No"
"169",35,1,"asymptomatic",126,282,0,2,156,1,0,1,0,"reversable","Yes"
"118",35,0,"asymptomatic",138,183,0,0,182,0,1.4,1,0,"normal","No"
"211",37,0,"nonanginal",120,215,0,0,170,0,0,1,0,"normal","No"
"4",37,1,"nonanginal",130,250,0,0,187,0,3.5,3,0,"normal","No"
"212",38,1,"typical",120,231,0,0,182,1,3.8,2,0,"reversable","Yes"
"303",38,1,"nonanginal",138,175,0,0,173,0,0,1,NA,"normal","No"
"223",39,0,"nonanginal",94,199,0,0,179,0,0,1,0,"normal","No"
"110",39,1,"asymptomatic",118,219,0,0,140,0,1.2,2,0,"reversable","Yes"
"278",39,0,"nonanginal",138,220,0,0,152,0,0,2,0,"normal","No"
"83",39,1,"nonanginal",140,321,0,2,182,0,0,1,0,"normal","No"
"30",40,1,"asymptomatic",110,167,0,2,114,1,2,2,0,"reversable","Yes"
"42",40,1,"typical",140,199,0,0,178,1,1.4,1,0,"reversable","No"
"269",40,1,"asymptomatic",152,223,0,0,181,0,0,1,0,"reversable","Yes"
"51",41,0,"nontypical",105,198,0,0,168,0,0,1,1,"normal","No"
"241",41,1,"nontypical",110,235,0,0,153,0,0,1,0,"normal","No"
"58",41,1,"asymptomatic",110,172,0,2,158,0,0,1,0,"reversable","Yes"
"148",41,1,"nonanginal",112,250,0,0,179,0,0,1,0,"normal","No"
"221",41,0,"nonanginal",112,268,0,2,172,1,0,1,0,"normal","No"
"296",41,1,"nontypical",120,157,0,0,182,0,0,1,0,"normal","No"
"242",41,0,"nontypical",126,306,0,0,163,0,0,1,0,"normal","No"
"213",41,1,"nonanginal",130,214,0,2,168,0,2,2,0,"normal","No"
"5",41,0,"nontypical",130,204,0,2,172,0,1.4,1,0,"normal","No"
"116",41,1,"nontypical",135,203,0,0,132,0,0,2,0,"fixed","No"
"152",42,0,"asymptomatic",102,265,0,2,122,0,0.6,2,0,"normal","No"
"187",42,1,"nonanginal",120,240,1,0,194,0,0.8,3,0,"reversable","No"
"240",42,1,"nontypical",120,295,0,0,162,0,0,1,0,"normal","No"
"256",42,0,"nonanginal",120,209,0,0,173,0,0,2,0,"normal","No"
"270",42,1,"nonanginal",130,180,0,0,150,0,0,1,0,"normal","No"
"266",42,1,"asymptomatic",136,315,0,0,125,1,1.8,2,0,"fixed","Yes"
"36",42,1,"asymptomatic",140,226,0,0,178,0,0,1,0,"normal","No"
"183",42,1,"typical",148,244,0,2,178,0,0.8,1,2,"normal","No"
"205",43,1,"asymptomatic",110,211,0,0,161,0,0,1,0,"reversable","No"
"255",43,1,"asymptomatic",115,303,0,0,181,0,1.2,2,0,"normal","No"
"37",43,1,"asymptomatic",120,177,0,2,120,1,2.5,2,0,"reversable","Yes"
"135",43,0,"nonanginal",122,213,0,0,165,0,0.2,2,0,"normal","No"
"179",43,1,"nonanginal",130,315,0,0,162,0,1.9,1,1,"normal","No"
"114",43,0,"asymptomatic",132,341,1,2,136,1,3,2,0,"reversable","Yes"
"193",43,1,"asymptomatic",132,247,1,2,143,1,0.1,2,NA,"reversable","Yes"
"29",43,1,"asymptomatic",150,247,0,0,171,0,1.5,1,0,"normal","No"
"94",44,0,"nonanginal",108,141,0,0,175,0,0.6,2,0,"normal","No"
"75",44,1,"asymptomatic",110,197,0,2,177,0,0,1,1,"normal","Yes"
"53",44,1,"asymptomatic",112,290,0,2,153,0,0,1,1,"normal","Yes"
"261",44,0,"nonanginal",118,242,0,0,149,0,0.3,2,1,"normal","No"
"129",44,1,"nontypical",120,220,0,0,170,0,0,1,0,"normal","No"
"14",44,1,"nontypical",120,263,0,0,173,0,0,1,0,"reversable","No"
"264",44,1,"nonanginal",120,226,0,0,169,0,0,1,0,"normal","No"
"293",44,1,"asymptomatic",120,169,0,0,144,1,2.8,3,0,"fixed","Yes"
"35",44,1,"nonanginal",130,233,0,0,179,1,0.4,1,0,"normal","No"
```

Sort command
-t is the field separator
‘,’ separator
-k key
2 ,5 column number to be sorted
2 age; 5 restbp
-n numeric
Heart filename

Jul 31 2025
BDBP106: Linux and Python programming

(vi) Sort the data according to sex, then according to age, then according to ChestPain. Explain the steps clearly.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ sort -t ',' -k3,3 -k2,2 -n -k4,4 Heart -o sexagechestpain_sorted
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ cat sexagechestpain_sorted
"","Age","Sex","ChestPain","RestBP","Chol","Fbs","RestECG","MaxHR","ExAng","Oldpeak","Slope","Ca","Thal","AHD"
"226",34,0,"nontypical",118,210,0,0,192,0,0.7,1,0,"normal","No"
"118",35,0,"asymptomatic",138,183,0,0,182,0,1.4,1,0,"normal","No"
"211",37,0,"nonanginal",120,215,0,0,170,0,0,1,0,"normal","No"
"223",39,0,"nonanginal",94,199,0,0,179,0,0,1,0,"normal","No"
"278",39,0,"nonanginal",138,220,0,0,152,0,0,2,0,"normal","No"
"221",41,0,"nonanginal",112,268,0,2,172,1,0,1,0,"normal","No"
"242",41,0,"nontypical",126,306,0,0,163,0,0,1,0,"normal","No"
"51",41,0,"nontypical",105,198,0,0,168,0,0,1,1,"normal","No"
"5",41,0,"nontypical",130,204,0,2,172,0,1.4,1,0,"normal","No"
"152",42,0,"asymptomatic",102,265,0,2,122,0,0.6,2,0,"normal","No"
"256",42,0,"nonanginal",120,209,0,0,173,0,0,2,0,"normal","No"
"114",43,0,"asymptomatic",132,341,1,2,136,1,3,2,0,"reversable","Yes"
"135",43,0,"nonanginal",122,213,0,0,165,0,0.2,2,0,"normal","No"
"261",44,0,"nonanginal",118,242,0,0,149,0,0.3,2,1,"normal","No"
"94",44,0,"nonanginal",108,141,0,0,175,0,0.6,2,0,"normal","No"
"126",45,0,"nontypical",130,234,0,2,175,0,0.6,2,0,"normal","No"
"170",45,0,"nontypical",112,160,0,0,138,0,0,2,0,"normal","No"
"198",45,0,"asymptomatic",138,236,0,2,152,1,0.2,2,0,"normal","No"
"217",46,0,"nontypical",105,204,0,0,172,0,0,1,0,"normal","No"
"218",46,0,"asymptomatic",138,243,0,2,152,1,0,2,0,"normal","No"
"62",46,0,"nonanginal",142,177,0,2,160,1,1.4,3,0,"normal","No"
"19",48,0,"nonanginal",130,275,0,0,139,0,0.2,1,0,"normal","No"
"239",49,0,"nontypical",134,271,0,0,162,0,0,2,0,"normal","No"
"243",49,0,"asymptomatic",130,269,0,0,163,0,0,1,0,"normal","No"
"199",50,0,"nontypical",120,244,0,0,162,0,1.1,1,0,"normal","No"
"201",50,0,"asymptomatic",110,254,0,2,159,0,0,1,0,"normal","No"
"26",50,0,"nonanginal",120,219,0,0,158,0,1.6,2,0,"normal","No"
"254",51,0,"nonanginal",120,295,0,2,157,0,0.6,1,0,"normal","No"
"61",51,0,"asymptomatic",130,305,0,0,142,1,1.2,2,0,"reversable","Yes"
"78",51,0,"nonanginal",140,308,0,2,142,0,1.5,1,1,"normal","No"
"90",51,0,"nonanginal",130,256,0,2,149,0,0.5,1,0,"normal","No"
"231",52,0,"nonanginal",136,196,0,2,169,0,0.1,2,0,"normal","No"
"82",53,0,"asymptomatic",130,264,0,2,143,0,0.4,2,0,"normal","No"
"88",53,0,"nonanginal",128,216,0,2,115,0,0,1,0,NA,"No"
"89",53,0,"asymptomatic",138,234,0,2,160,0,0,1,0,"normal","No"
"163",54,0,"nonanginal",110,214,0,0,158,0,1.6,2,0,"normal","No"
"168",54,0,"nontypical",132,288,1,2,159,1,0,1,1,"normal","No"
"222",54,0,"nonanginal",108,267,0,2,167,0,0,1,0,"normal","No"
"235",54,0,"nonanginal",160,201,0,0,163,0,0,1,1,"normal","No"
"64",54,0,"nonanginal",135,304,1,0,170,0,0,1,0,"normal","No"
"136",55,0,"nontypical",135,250,0,2,161,0,1.4,2,0,"normal","No"
"232",55,0,"asymptomatic",180,327,0,1,117,1,3.4,2,0,"normal","Yes"
"283",55,0,"asymptomatic",128,205,0,1,130,1,2,2,1,"reversable","Yes"
"292",55,0,"nontypical",132,342,0,0,166,0,1.2,1,0,"normal","No"
"12",56,0,"nontypical",140,294,0,2,153,0,1.3,2,0,"normal","No"
"127",56,0,"asymptomatic",200,288,1,2,133,1,4,3,2,"reversable","Yes"
"182",56,0,"asymptomatic",134,409,0,2,150,1,1.9,2,2,"reversable","Yes"
"103",57,0,"asymptomatic",128,303,0,2,159,0,0,1,1,"normal","No"
"298",57,0,"asymptomatic",140,241,0,0,123,1,0.2,2,0,"reversable","Yes"
"302",57,0,"nontypical",130,236,0,2,174,0,0,2,1,"normal","Yes"
"8",57,0,"asymptomatic",120,354,0,0,163,1,0.6,1,0,"normal","No"
"164",58,0,"asymptomatic",100,248,0,2,122,0,1,2,0,"normal","No"
"22",58,0,"typical",150,283,1,2,162,0,1,1,0,"normal","No"
```

Sort command
-t is the field separator
‘,’ separator
-k key
3,2 ,4 column number to be sorted
2 age; 3 sex; 4 chestpain
-n numeric
Heart filename

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ ls -li
total 404
28579245 -rw-rw-r--  1 ibab ibab 19925 Jul 31 16:48 agechestpain_sorted
28578778 -rw-rw-r--  1 ibab ibab 19925 Jul 31 15:45 age_sorted.out
28578757 -rw-rw-r--  1 ibab ibab 19925 Jul 31 15:42 col1_sorted.out
28594288 drwxrwxrwx 83 ibab ibab  4096 Jul 22 16:59 etc_learn
28596577 -rw-rw-r--  1 ibab ibab 96755 Jul 24 14:34 etc.zip
28603737 -rw-rw-r--  1 ibab ibab    24 Jul 31 09:30 file1
28603773 -rw-rw-r--  1 ibab ibab    23 Jul 31 09:31 file2
28603777 -rw-rw-r--  1 ibab ibab    48 Jul 31 09:31 file3
28603918 -rw-rw-r--  1 ibab ibab    48 Jul 31 09:39 file3_sorted
28577417 -rw-rw-r--  3 ibab ibab 19925 Jul 31 14:00 Hardheart
28577417 -rw-rw-r--  3 ibab ibab 19925 Jul 31 14:00 Heart
28579064 -rw-rw-r--  1 ibab ibab 19318 Jul 31 16:17 Heart.csv
28588328 -rw-rw-r--  1 ibab ibab 47489 Jul 18 15:04 lab2.pdf
28577365 -rw-rw-r--  1 ibab ibab 57441 Jul 31 14:00 lab4.pdf
28578787 -rw-rw-r--  1 ibab ibab 19925 Jul 31 15:47 restbp_revsort.out
28579257 -rw-rw-r--  1 ibab ibab 19925 Jul 31 16:51 sexagechestpain_sorted
28578876 -rw-rw-r--  1 ibab ibab 19925 Jul 31 16:31 sex_sorted
28577987 lrwxrwxrwx  1 ibab ibab     8 Jul 31 14:25 symheart -> symheart
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$
```